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2. Finite State Machine (Pre requisites)

1. Symbol - a, b, c, 0, 1, 2, 3, ...
2. Alphbet - ~~A~~ Σ - collection of symbols.
   Eg. $\{a, b\}$, $\{d, e, f, g\}$
3. String - sequence of symbols
   Eg: a, b, 0, 1, aa, bb, ab, 01, ...
4. Language - Set of strings
   Eg. $\Sigma = \{0, 1\}$

Finite
L_1 = set of all strings of length 2
$= \{00, 01, 10, 11\}$
L_2 = set of all strings of length 3
$= \{000, 001, 010, 011, 100, 101, 110, 111\}$

L_3 = set of all strings that begin with 0.
$= \{0, 00, 01, 000, 001, 010, 011, 0000, \ldots\}$
infinite

5. Powers of Σ: suppose $\Sigma = \{0, 1\}$
Σ^0 = set of all strings of length 0

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$\Rightarrow \Sigma^0 = \{\varepsilon\}$ → Epslion

Σ^1 = set of all strings of length 1:

$\Sigma^1 = \{0, 1\}$

Σ^2 = 〃 length 2. $\Sigma^2 = \{00, 01, 10, 11\}$

Σ^3 = set of all strings of length 3 〃

⋮

Σ^n = set of all strings of length n

6, Cardinality: number of elements in a set.

for Σ^0 is 1, Σ^1 is 2, Σ^2 is 4

$\Rightarrow \Sigma^n = 2^n$ for $\Sigma = \{0, 1\}$

7, $\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \Sigma^4 \cdots$

$= \{\varepsilon\} \cup \{0, 1\} \cup \{00, 01, 10, 11\} \cup \cdots$

= Set of all possible strings of all lengths over $\{0, 1\}$